Fundamentals of AI and KR - Module 3

7. Exercises: First part

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Exercise 1

Exercise 1

Which positive test is more informative?

Scenario

A virus is present in 1% of the population. A randomly selected person takes one diagnostic test.

Two tests are available:

	Test A	**Test B**
Sensitivity: $P(+ \mid D)$	0.95	0.90
False positive rate: $P(+ \mid \neg D)$	0.10	0.05

D: the person has the virus.

$+$: the test result is positive.

Question

Which test, when positive, gives a higher probability that the person actually has the virus?

Adapted from Russell and Norvig, AIMA online exercises, Exercise 13.16.

Your turn: identify and calculate

1. Before calculating, which positive result would you consider stronger evidence of infection? Why?
2. Identify the prior probabilities:
$$P(D), \qquad P(\neg D).$$
3. Calculate the probability of a positive result for each test:
$$P(+_A), \qquad P(+_B).$$
4. Calculate and compare the posterior probabilities:
$$P(D \mid +_A), \qquad P(D \mid +_B).$$

Hint

A positive result may be a true positive or a false positive. Account for both possibilities.

Step 1: how likely is a positive result?

2. Identify the prior probabilities: $P(D)$, $P(\neg D)$.

The prior probabilities are: $P(D) = 0.01$, $P(\neg D) = 0.99$.

3. Calculate the probability of a positive result for each test: $P(+_A), P(+_B)$.

Positive result can occur in two mutually exclusive cases, for total probability law:

$$P(+) = P(+ \mid D)P(D) + P(+ \mid \neg D)P(\neg D).$$

Test A

$$\begin{aligned} P(+_A) &= 0.95 \cdot 0.01 + 0.10 \cdot 0.99 \\ &= 0.0095 + 0.099 = 0.1085. \end{aligned}$$

Test B

$$\begin{aligned} P(+_B) &= 0.90 \cdot 0.01 + 0.05 \cdot 0.99 \\ &= 0.009 + 0.0495 = 0.0585. \end{aligned}$$

Step 2: update the probability of infection

4. Calculate and compare the posterior probabilities: $P(D \mid +_A), P(D \mid +_B)$.

Bayes' rule reverses the direction of conditioning:

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}.$$

Test A

$$P(D \mid +_A) = \frac{0.95 \cdot 0.01}{0.1085} \approx 0.0876 = 8.76\%.$$

Test B

$$P(D \mid +_B) = \frac{0.90 \cdot 0.01}{0.0585} \approx 0.1538 = 15.38\%.$$

A positive result from **Test B** gives a higher posterior probability of infection, despite its lower sensitivity.

Why? Imagine testing 10,000 people

With a prevalence of 1%, we expect:

100 infected people, 9,900 uninfected people.

Expected counts	**Test A**	**Test B**
True positives	95	90
False positives	990	495
Total positives	1,085	585

Among those who test positive, the expected infected fractions are:

$$\text{Test A: } \frac{95}{1{,}085} \approx 8.76\%, \qquad \text{Test B: } \frac{90}{585} \approx 15.38\%.$$

Interpretation

Test B detects fewer infected people, but produces far fewer false positives. Larger fraction of its positive results therefore corresponds to infected people.

Exercise 2

Exercise 2

Which coin was selected?

Scenario

A bag contains three biased coins:

- coin a: probability of heads 0.20;
- coin b: probability of heads 0.60;
- coin c: probability of heads 0.80.

One coin is selected uniformly at random and tossed three times. The same coin is used for all three tosses.

Let $C \in \{a, b, c\}$ identify the selected coin, and let $X_1, X_2, X_3 \in \{H, T\}$ denote the outcomes.

Question

We observe: $X_1 = H$, $X_2 = H$, $X_3 = T$. Which coin is most likely to have been selected?

Adapted from Russell and Norvig, AIMA online exercises, Exercise 14.1.

Your turn: build the model and answer the query

1. Draw a Bayesian network for C, X_1, X_2, X_3.
2. Specify the prior distribution of C and the CPT for each toss.
3. Write the factorization of the joint distribution:

$$P(C, X_1, X_2, X_3).$$

4. Calculate the posterior distribution:

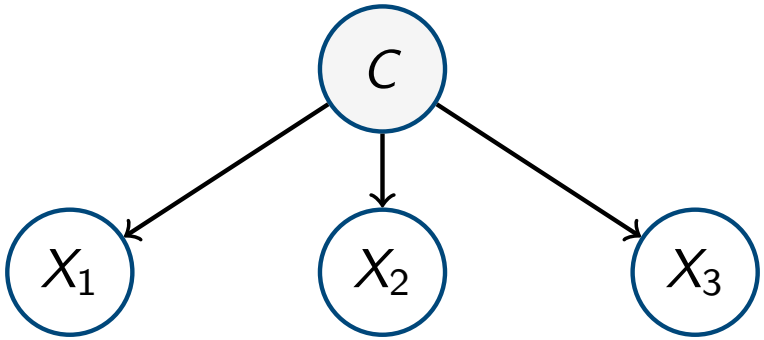
$$P(C \mid X_1 = H, X_2 = H, X_3 = T).$$

5. Are the toss outcomes independent:
   - when the selected coin is known?
   - when the selected coin is unknown?

Step 1: represent the hidden common cause

1. Draw a Bayesian network for C, X_1, X_2, X_3.

The selected coin determines the probability of heads on each toss.



For a given coin, successive tosses are independent. Thus, the outcomes are conditionally independent given C.

3. Write the factorization of the joint distribution: $P(C, X_1, X_2, X_3)$.

The joint distribution factorizes as:

$$P(C, X_1, X_2, X_3) = P(C) \prod_{i=1}^{3} P(X_i \mid C).$$

Step 2: specify the local distributions

2. Specify the prior distribution of C and the CPT for each toss.

Each coin is equally likely to be selected:

$$P(C = a) = P(C = b) = P(C = c) = \frac{1}{3}.$$

Each toss has the same conditional probability table:

Selected coin	$P(X_i = H \mid C)$	$P(X_i = T \mid C)$
a	0.20	0.80
b	0.60	0.40
c	0.80	0.20

For example, if coin b was selected, every toss has probability 0.60 of heads and 0.40 of tails. The three toss nodes share the same CPT.

Step 3: how well does each coin explain the observations?

4. Calculate the posterior distribution: $P(C \mid X_1 = H, X_2 = H, X_3 = T)$.

Conditional independence gives:

$$P(e \mid C) = P(X_1 = H \mid C)\, P(X_2 = H \mid C)\, P(X_3 = T \mid C).$$

Coin	**Calculation**	**Likelihood** $P(e \mid C)$
a	$0.20^2 \cdot 0.80$	0.032
b	$0.60^2 \cdot 0.40$	0.144
c	$0.80^2 \cdot 0.20$	0.128

Interpretation

Coin b gives the observed sequence the highest likelihood.
Coin c makes the two heads more likely, but makes the tail less likely. We must account for **all three observations**.

Step 4: apply Bayes' rule and normalize

For each coin value $k \in \{a, b, c\}$:

$$P(C = k \mid e) = \frac{P(e \mid C = k)P(C = k)}{\sum_{u \in \{a,b,c\}} P(e \mid C = u)P(C = u)} =$$

$$= \frac{P(e \mid C = k) \cdot \frac{1}{3}}{\frac{1}{3}\sum_{u \in \{a,b,c\}} P(e \mid C = u)} = \frac{P(e \mid C = k)}{\sum_{u \in \{a,b,c\}} P(e \mid C = u)}.$$

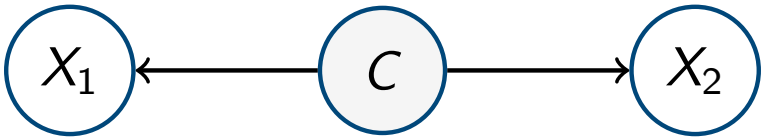
Thus, we divide each likelihood by their sum: $0.032 + 0.144 + 0.128 = 0.304$.

Coin	Posterior probability	Approximate value
a	$0.032/0.304$	10.53%
b	$0.144/0.304$	47.37%
c	$0.128/0.304$	42.11%

Coin b is the most probable explanation.

Independence: a non-collider on the trail

5. Are the toss outcomes independent:
   - when the selected coin is known?
   - when the selected coin is unknown?



The selected coin C is a common cause. It is a non-collider on this trail.

C unobserved: active trail

Observing heads on the first toss changes our belief about which coin was selected. This changes our prediction for the second toss.

C observed: blocked trail

Observing a non-collider blocks the trail. Once the coin is known, the first outcome adds no information about the second.

Are the toss outcomes independent?

When the coin is known: yes

Knowing one outcome gives no additional information about another:

$$P(X_2 = H \mid X_1 = H, C = b) = P(X_2 = H \mid C = b) = 0.60.$$

When the coin is unknown: no

Before observing any toss:

$$P(X_2 = H) = \frac{0.20 + 0.60 + 0.80}{3} = \frac{8}{15} \approx 0.533.$$

After observing heads on the first toss:

$$P(X_2 = H \mid X_1 = H) = \frac{(0.20^2 + 0.60^2 + 0.80^2)/3}{(0.20 + 0.60 + 0.80)/3}$$
$$= 0.65.$$

Exercise 3

Exercise 3

Can we trust the alarm?

Scenario

A nuclear power plant uses a temperature gauge to monitor its core. An alarm sounds when the gauge reports a high temperature. Both the gauge and the alarm can fail. The gauge is more likely to fail when the actual core temperature is high.

We consider five variables:

- T: actual core temperature;
- G: temperature reported by the gauge;
- F_G: the gauge is faulty;
- F_A: the alarm is faulty;
- A: the alarm sounds.

For this exercise, $T, G \in \{normal, high\}$. The other variables are Boolean.

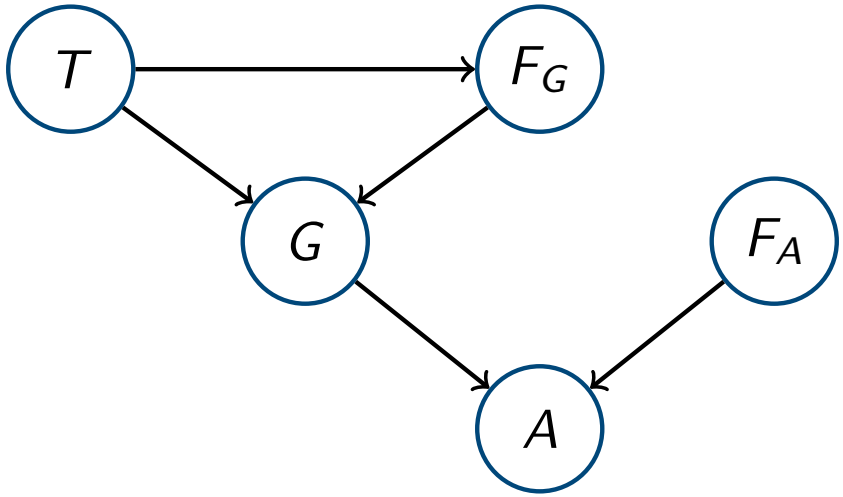
Adapted from Russell and Norvig, AIMA online exercises, Exercise 14.13.

Your turn: build and analyse the network

1. Draw a Bayesian network and motivate each arrow.
2. Write the factorization of the joint distribution.
3. Specify the CPT for G, assuming that the probability of a correct reading is:
   - x when the gauge is working;
   - y when the gauge is faulty.
4. Specify the CPT for A: a working alarm sounds exactly when the gauge reports *high*; a faulty alarm never sounds.
5. Count the free parameters of the network with generic CPTs.
6. Identify the graphical independences involving G and determine its Markov blanket.
7. Identify the graphical independences involving A and determine its Markov blanket.

Step 1: choose the direct dependencies

1. Draw a Bayesian network and motivate each arrow.



- $T \to F_G$: high temperature increases the risk of gauge failure.
- $T \to G$, $F_G \to G$: the reading depends on temperature and gauge condition.
- $G \to A$, $F_A \to A$: the alarm depends on the reading and alarm condition.

We assume that the alarm's fault status F_A is independent of T and F_G.

Step 2: write the joint distribution

2. Write the factorization of the joint distribution.

Each node contributes its probability conditional on its parents:

Node	**Parents**	**Local model**
T	none	$P(T)$
F_A	none	$P(F_A)$
F_G	T	$P(F_G \mid T)$
G	T, F_G	$P(G \mid T, F_G)$
A	G, F_A	$P(A \mid G, F_A)$

Therefore:

$$P(T, F_G, G, F_A, A) = P(T)\, P(F_A)\, P(F_G \mid T) \cdot P(G \mid T, F_G)\, P(A \mid G, F_A).$$

Step 3: specify the gauge's CPT

3. Specify the CPT for G, assuming that the probability of a correct reading is:
   - x when the gauge is working;
   - y when the gauge is faulty.

Parent values		**Gauge reading**	
T	F_G	$P(G = high \mid T, F_G)$	$P(G = normal \mid T, F_G)$
high	*false*	x	$1 - x$
normal	*false*	$1 - x$	x
high	*true*	y	$1 - y$
normal	*true*	$1 - y$	y

Step 4: specify the alarm's CPT

4. Specify the CPT for A: a working alarm sounds exactly when the gauge reports *high*; a faulty alarm never sounds.

Parent values		**Alarm state**	
G	F_A	$P(A = true \mid G, F_A)$	$P(A = false \mid G, F_A)$
high	*false*	1	0
normal	*false*	0	1
high	*true*	0	1
normal	*true*	0	1

A deterministic local model

Once G and F_A are known, the alarm state is determined:

$$A = true \iff (G = high) \wedge (F_A = false).$$

Uncertainty remains because the parent values may be unknown.

Step 5: count the free parameters

5. Count the free parameters of the network with generic CPTs.

All 5 variables have two possible values.

Local distribution	**Free parameters**
$P(T)$	1
$P(F_A)$	1
$P(F_G \mid T)$	2
$P(G \mid T, F_G)$	4
$P(A \mid G, F_A)$	4
Total	**12**

Remark: The exercise imposes additional restrictions

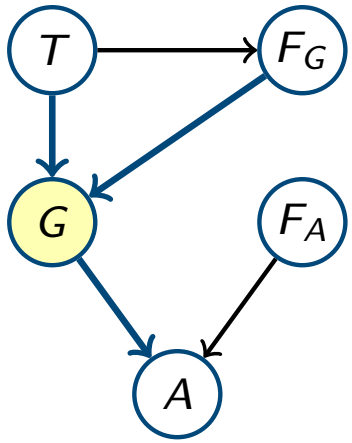
The gauge's four rows use only x, y, and the alarm's CPT is fixed. With these restrictions, the model has: $1 + 1 + 2 + 2 + 0 = 6$ free numerical parameters.

Step 6: read independence from the graph

6. Identify the graphical independences involving G and determine its Markov blanket.

First, consider the nodes directly connected to G:

- its parents: T, F_G;
- its child: A.



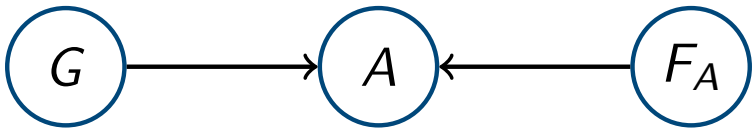
What does the graph tell us?

The graph does not imply independence between G and any of these three adjacent nodes.

The remaining node to examine is F_A.

Step 6: G and F_A without evidence

The only trail between G and F_A is:



The arrows meet at A, so A is a collider.

Unobserved collider: blocked trail

Since A and its descendants are unobserved, the trail is **blocked**:

$$\boxed{G \perp F_A.}$$

Interpretation

Without observing the alarm, learning whether it is faulty provides no information about the gauge reading:

$$P(G \mid F_A) = P(G).$$

Step 6: independence given the parents of G

The local semantics states that a node is independent of its other non-descendants, given its parents.

Apply local semantics

$$\boxed{G \perp F_A \mid T, F_G.}$$

Equivalently:

$$P(G \mid T, F_G, F_A) = P(G \mid T, F_G).$$

Once the actual temperature and gauge condition are known, learning the alarm's fault status adds no information about G. This statement does not include the descendant A.

Step 6: which observations preserve independence?

The only trail between G and F_A is:

$$G \rightarrow A \leftarrow F_A.$$

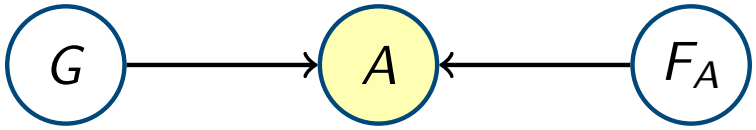
It remains **blocked** as long as the collider A is unobserved. Here, A has no descendants.

Observed variables	**Independence implied by the graph**
None	$G \perp F_A$
T	$G \perp F_A \mid T$
F_G	$G \perp F_A \mid F_G$
T, F_G	$G \perp F_A \mid T, F_G$

Why do all four statements hold?

Neither T nor F_G lies on this trail, and neither is a descendant of its collider A. Observing them therefore does not open the trail. The last statement also follows directly from local semantics.

Step 6: what changes when A is observed?



A observed

Observing the collider A makes the trail **active**. The graph no longer guarantees independence between G and F_A.

Suppose the alarm is silent: $A = \mathit{false}$

Silence may occur because the gauge reports *normal*, or because the alarm is faulty.

If we now learn that the gauge reports *high*, the alarm must be faulty under our model:

$$P(F_A = \mathit{true} \mid A = \mathit{false}, G = \mathit{high}) = 1.$$

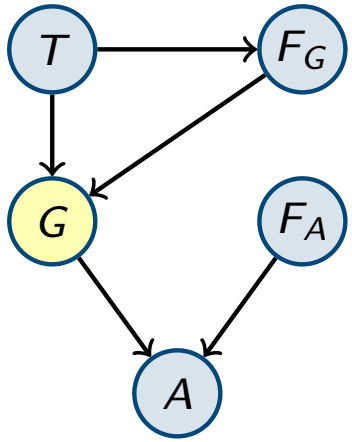
The gauge reading can therefore help us interpret the observed silence.

Step 6: the Markov blanket of G

The Markov blanket contains:

- parents: T, F_G;
- children: A;
- other parents of its children: F_A.

$$\boxed{MB(G) = \{T, F_G, A, F_A\}.}$$



Here, every other node belongs to the Markov blanket of G.

Why include F_A if $G \perp F_A$?

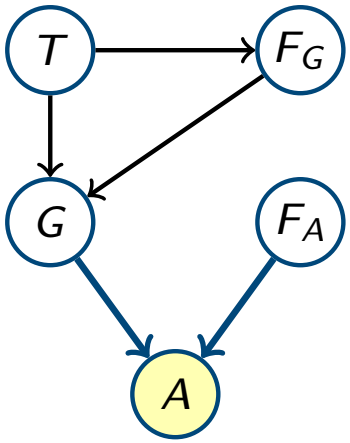
The independence holds when the alarm is unobserved.
The Markov blanket also includes the child A. Once A is observed, its other parent F_A can help us interpret that observation as evidence about G.

Step 7: read independence from the graph

7. Identify the graphical independences involving A and determine its Markov blanket.

First, identify the nodes directly connected to A:

- parents: G, F_A;
- children: none.



Direct connections

The graph does not imply independence between A and either parent.

The remaining nodes to examine are T and F_G.

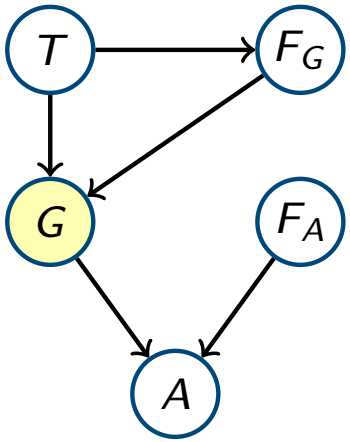
Step 7: which trails connect A to its ancestors?

The trails between T and A are:

$$T \to G \to A,$$
$$T \to F_G \to G \to A.$$

The trails between F_G and A are:

$$F_G \to G \to A,$$
$$F_G \leftarrow T \to G \to A.$$



G observed

Every trail passes through G, which is a non-collider.

Observing G blocks every trail

$$\boxed{A \perp \{T, F_G\} \mid G.}$$

Once the gauge reading is known, the actual temperature and gauge condition add no information about whether the alarm sounds.

Step 7: why observing other variables is not enough

To separate A from T

Observing F_G blocks: $T \rightarrow F_G \rightarrow G \rightarrow A$. But the shorter trail remains active:

$$T \rightarrow G \rightarrow A.$$

Therefore, observing F_G alone does not d-separate A and T.

To separate A from F_G

Observing T blocks: $F_G \leftarrow T \rightarrow G \rightarrow A$. But the shorter trail remains active:

$$F_G \rightarrow G \rightarrow A.$$

Therefore, observing T alone does not d-separate A and F_G.

Step 7: apply local semantics to A

For A:

- parents: G, F_A;
- descendants: none;
- other non-descendants: T, F_G.

Local semantics

A node is independent of its other non-descendants, given its parents:

$$\boxed{A \perp \{T, F_G\} \mid G, F_A.}$$

Equivalently:

$$P(A \mid G, F_A, T, F_G) = P(A \mid G, F_A).$$

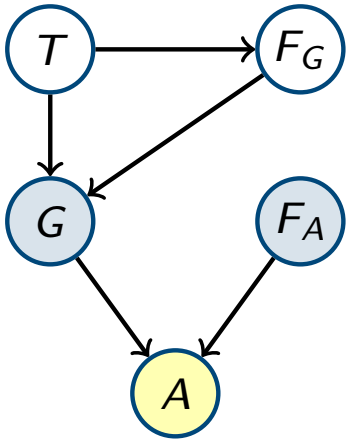
Once we know the gauge reading and whether the alarm is faulty, the actual temperature and gauge condition provide no additional information about the alarm state.

Step 7: the Markov blanket of A

The Markov blanket contains:

- parents: G, F_A;
- children: none;
- other parents of children: none.

$$\boxed{MB(A) = \{G, F_A\}.}$$



The variables outside the blanket are T, F_G.

Why do we need both parents?

Knowing G makes T, F_G irrelevant for predicting A, but F_A can still change that prediction.
Knowing both G and F_A allows us to ignore every other node.

Step 7: compare local semantics and Markov blanket

Local semantics

Given its parents, A is independent of its other non-descendants:

$$A \perp \{T, F_G\} \mid G, F_A.$$

Markov blanket

Given its blanket, A is independent of all nodes outside it:

$$A \perp \{T, F_G\} \mid G, F_A.$$

Why are the statements identical here?

A has no children or other descendants.
For a node with no children, the Markov blanket consists exactly of its parents.

Exercise 4

Exercise 4

Why is the service unavailable?

Scenario

An online service may become unavailable because of:

- a power problem, P;
- a network problem, N;
- an overload, O.

Each problem can make the service unavailable, but does not necessarily do so: backup mechanisms may prevent the outage. Let $U = \mathit{true}$ mean that the service is unavailable. All four variables are Boolean.

We use a Noisy-OR without leak, with inhibition probabilities:

$$q_P = 0.10, \qquad q_N = 0.20, \qquad q_O = 0.40.$$

Conditional on active causes, their mechanisms fail independently to trigger U.

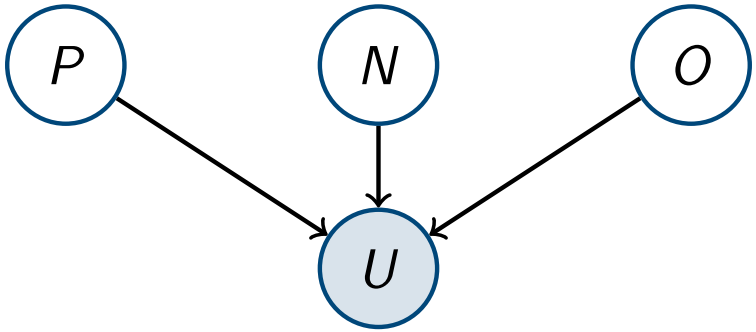
Application of Noisy-OR, inspired by AIMA Exercise 14.10.

Your turn: combine the causes

1. Draw the Bayesian network.
2. Explain the meaning of $q_N = 0.20$.
3. Calculate:
$$P(U = true \mid P = true, N = true, O = false),$$
$$P(U = true \mid P = true, N = true, O = true).$$
4. What is the probability of unavailability when all three causes are false?
5. Compare the number of free parameters required by a generic CPT and by this Noisy-OR model.
6. Does Noisy-OR require the causes P, N, O to be independent?

Step 1: draw the network and interpret q_i

1. Draw the Bayesian network.



2. Explain the meaning of $q_N = 0.20$.

When a network problem is present, its mechanism has probability 0.20 of failing to make the service unavailable. Probability of successfully trigger it is:

$$p_N = 1 - q_N = 0.80.$$

i.e., with no leak and no other active causes:

$$P(U = true \mid P = false, N = true, O = false) = 0.80.$$

Step 2: two active causes

3. Calculate:

$$P(U = \mathit{true} \mid P = \mathit{true}, N = \mathit{true}, O = \mathit{false}),$$

Both active mechanisms must fail to trigger unavailability. By the Noisy-OR:

$$\begin{aligned} &P(U = \mathit{false} \mid P = \mathit{true}, N = \mathit{true}, O = \mathit{false}) \\ &= q_P q_N = 0.10 \cdot 0.20 = 0.02. \end{aligned}$$

Therefore:

$$\begin{aligned} &P(U = \mathit{true} \mid P = \mathit{true}, N = \mathit{true}, O = \mathit{false}) \\ &= 1 - q_P q_N = 1 - 0.02 = \boxed{0.98}. \end{aligned}$$

The inactive cause O does not contribute to the product.

Step 3: all causes active, or no causes active I

3. Calculate:

$$P(U = true \mid P = true, N = true, O = true).$$

All three causes are active

The service remains available only if all three mechanisms fail to trigger unavailability:

$$\begin{aligned} &P(U = true \mid P = true, N = true, O = true) \\ &\quad = 1 - q_P q_N q_O \\ &\quad = 1 - (0.10 \cdot 0.20 \cdot 0.40) \\ &\quad = \boxed{0.992}. \end{aligned}$$

Step 3: all causes active, or no causes active II

4. What is the probability of unavailability when all three causes are false?

All three causes are inactive

Without leak, there is no other mechanism that can trigger U:

$$P(U = \mathit{true} \mid P = \mathit{false}, N = \mathit{false}, O = \mathit{false}) = \boxed{0}.$$

What if other causes are possible?

A leak parameter can represent unmodelled causes, allowing unavailability even when P, N, O are all false.

Step 4: three parameters determine the whole CPT

5. Compare the number of free parameters required by a generic CPT and by this Noisy-OR model.

P	N	O	$P(U = true \mid P, N, O)$	**Value**
false	*false*	*false*	0	0
true	*false*	*false*	$1 - q_P$	0.90
false	*true*	*false*	$1 - q_N$	0.80
false	*false*	*true*	$1 - q_O$	0.60
true	*true*	*false*	$1 - q_P q_N$	0.98
true	*false*	*true*	$1 - q_P q_O$	0.96
false	*true*	*true*	$1 - q_N q_O$	0.92
true	*true*	*true*	$1 - q_P q_N q_O$	0.992

A generic CPT for a Boolean child with three Boolean parents requires $2^3 = 8$ free parameters. The Noisy-OR without leak requires only 3: q_P, q_N, q_O.

The comparison concerns only the local distribution of U.

Step 5: what is assumed to be independent?

6. Does Noisy-OR require the causes P, N, O to be independent?

The independence assumption

Conditional on the active causes, the failures of their triggering mechanisms are independent.
This allows us to multiply inhibition probabilities:

$$P(U = \mathit{false} \mid P = \mathit{true}, N = \mathit{true}, O = \mathit{false}) = q_P q_N.$$

Must the causes themselves be independent?

No. For example, a power problem might also cause a network problem.
That dependency belongs in the model of the parent variables. It does not prevent us from specifying a Noisy-OR for:

$$P(U \mid P, N, O),$$

provided the mechanism assumptions remain appropriate.

Questions?

